

Wenjie Luo

AI Inference and Model Evaluation Engineer | Graduating Jan 2027

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PROFILE

Evaluate inference and model behavior with controlled experiments, numerical checks and scene-level analysis. Work with SGLang/vLLM, GPU kernels and PyTorch training.

EDUCATION

National University of Singapore | MSc Civil Engineering (Transportation) Aug 2025 - Jan 2027

Transport focus; research in transportation information and control.

Chang'an University | BEng Transportation Engineering Sep 2021 - Jun 2025

GPA 3.16/5.0; approximately top 15% of the cohort.

PROJECTS

SGLang/vLLM Serving and Performance Analysis Sep 2026

- On a T4 with Qwen2.5-0.5B, traced SGLang's throughput drop to decode-graph capture stopping at batch 8; tested causality by disabling graphs and extending capture.
- Raised the capture limit to 32: batch-32 throughput rose from 608 to 2,690 tok/s (4.42×) in the same experiment. Logged parameters and one measurement per batch.

fp16x3 Transformer GPU Optimization Aug 23, 2026 - Present

- Wrote Triton kernels to split operands into fp16 high/low components for one expanded-K cuBLAS Tensor Core GEMM; used fused attention and chunking to reduce memory use.
- T4: 13/13 FP32 shapes passed (max abs error 9.5e-6); 3 independent runs/shape gave 2.83× median speedup over the FP32 reference. A separate FP16 100k run peaked at 14.2 GB.

NAVSIM | Planning Model Training and Evaluation Aug - Sep 2026

- Trained MLP planners on 103,288 navtrain scenes and evaluated 12,146 navtest scenes; used a 2×2 map/agent-input design, three seeds per cell and paired 95% confidence intervals.
- Reproduced PDMS baselines: ConstantVelocity 20.7 (paper 20.6), Human 94.6 (94.8). Ground-truth boxes significantly lowered map-aware PDMS across three seeds despite similar L1.

EXPERIENCE

Far East Facade | Project Engineer, Engineering Department Jun 2026 - Present

- Used Python to extract delivery batches and test results from supplier emails, supporting status tracking for 15 façade material types.

Caixun Technology | Product Intern Jun 2025 - Aug 2025

- Moved the hard-to-find prize-draw entry from the page bottom to a side pop-up to improve visibility; defined requirements, designed interaction prototypes and followed development and testing.
- Independently wrote 3 PRDs and prototyped 5 core pages in Axure; proposals were accepted and scheduled for development. Tracked fixes and retesting, helping resolve over 10 issues.

SKILLS

Inference: SGLang, vLLM, CUDA C++, Triton, PyTorch, CUDA Graphs

Evaluation: controlled experiments, numerical checks, paired bootstrap, factorial designs, multiple seeds